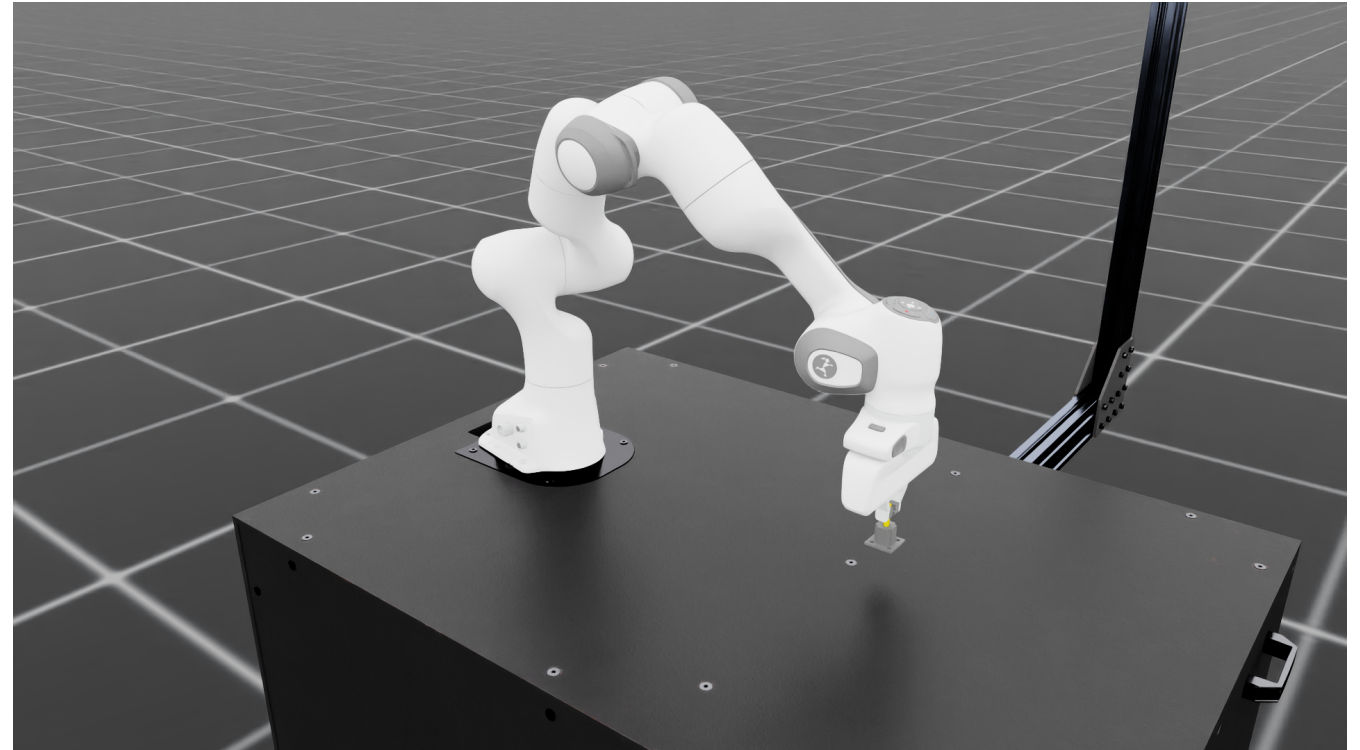


# Preventing insertion-induced jamming

First avoid risky insertion.  
Then restore a force-limited exit.

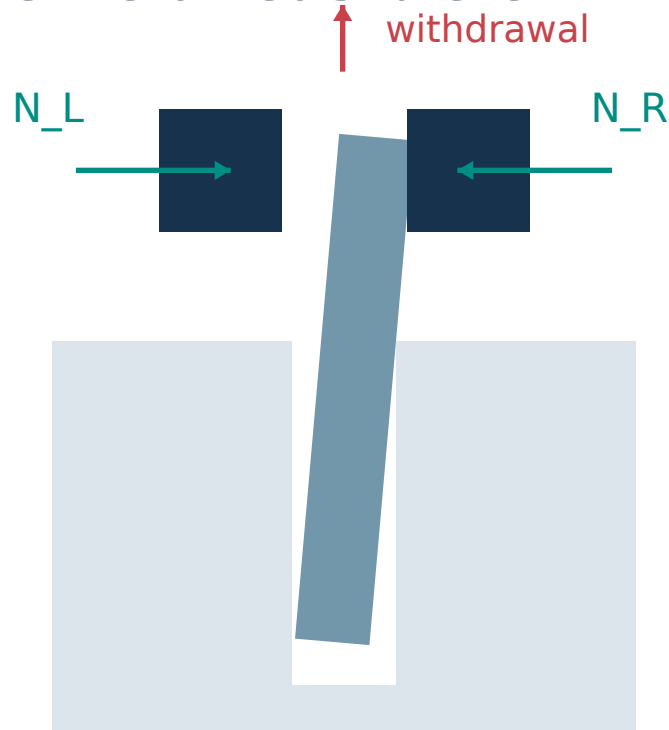
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Research progress and technical roadmap  
September 2026



# Prevention first, recovery second

## Preserve a feasible exit throughout insertion



Peg and socket cutaway  
Schematic, clearance enlarged

### 01 Avoid insertion-induced jamming

Choose pose, speed and compliance before a risky contact state develops.

### 02 Restore withdrawability

If progress is blocked, adjust posture and withdraw within the operation-force budget.

### Why bounded force?

Limited grip and part strength motivate the constraint.  
Exploratory wrist budgets: 3 / 4 / 5 N, torque  $\leq 1$  N·m.

# Research context

|                                 |                                                                    |                                           |
|---------------------------------|--------------------------------------------------------------------|-------------------------------------------|
| <b>Contact and extraction</b>   | Geometry, friction and compliance shape mating forces.             | Whitney 1982 [1]<br>Goli et al. 2024 [2]  |
| <b>Interaction control</b>      | Virtual stiffness and damping shape contact response.              | Hogan 1985 [3]                            |
| <b>Planning through contact</b> | Trajectory optimization can include changing contact modes.        | Posa et al. 2014 [4]                      |
| <b>Safety and transfer</b>      | Predictive filtering and calibrated deployment address model risk. | Wabersich 2021 [8]<br>IndustReal 2023 [6] |

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**First avoid risky insertion. Then restore an exit within the force budget.**

# Plan insertion to preserve a feasible exit



## Kinematics and interaction control

$$\tau = J(q)^T (K_p e - K_d \dot{x}) + \tau_{null}$$

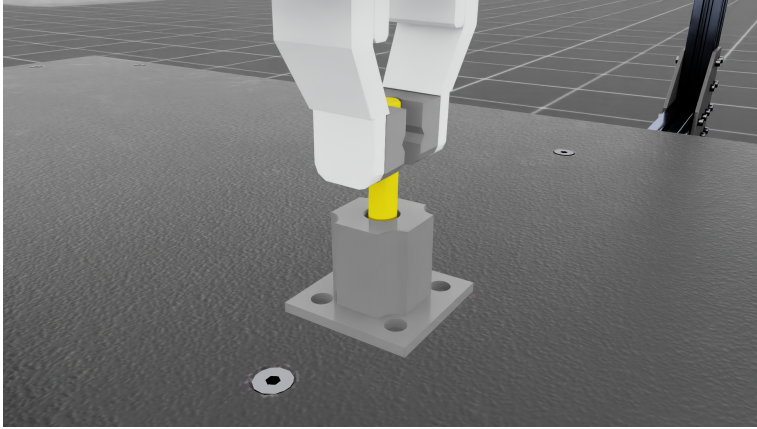
Existing task-space PD law, simplified.  $e$  is the position/orientation error.

**Primary objective** Advance insertion while avoiding jam-risk states and retaining a predicted feasible exit.

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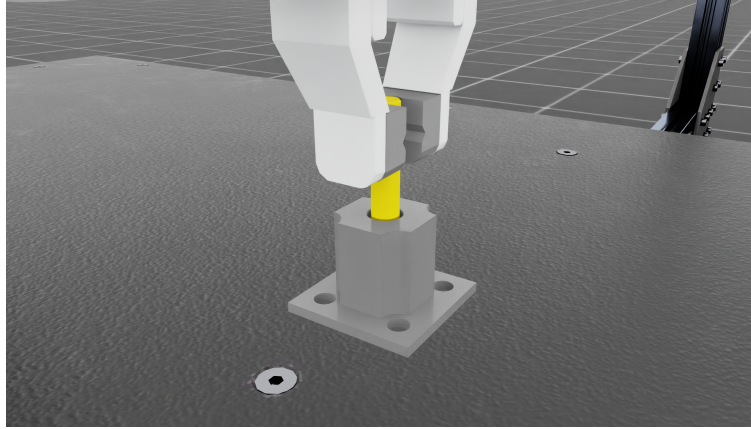
**If progress is blocked: restore an exit within the force budget.**

# Current testbed and recovery probes



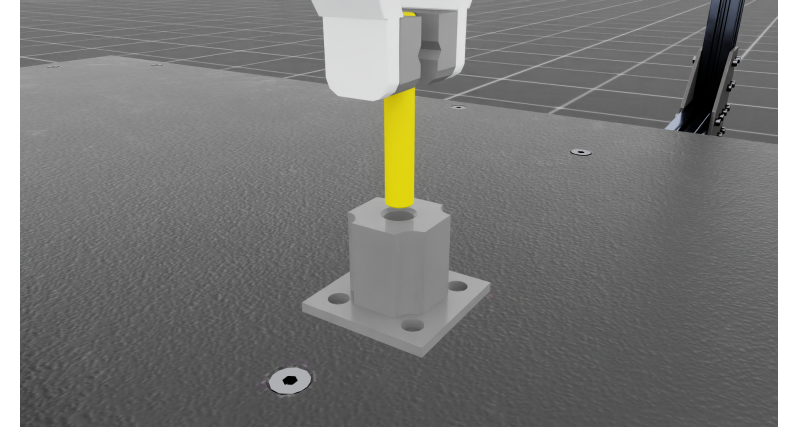
## Reference state

18.228 mm depth, 1.439° tilt



## Recentered and realigned

18.076 mm depth, 0.170° tilt



## Clearance

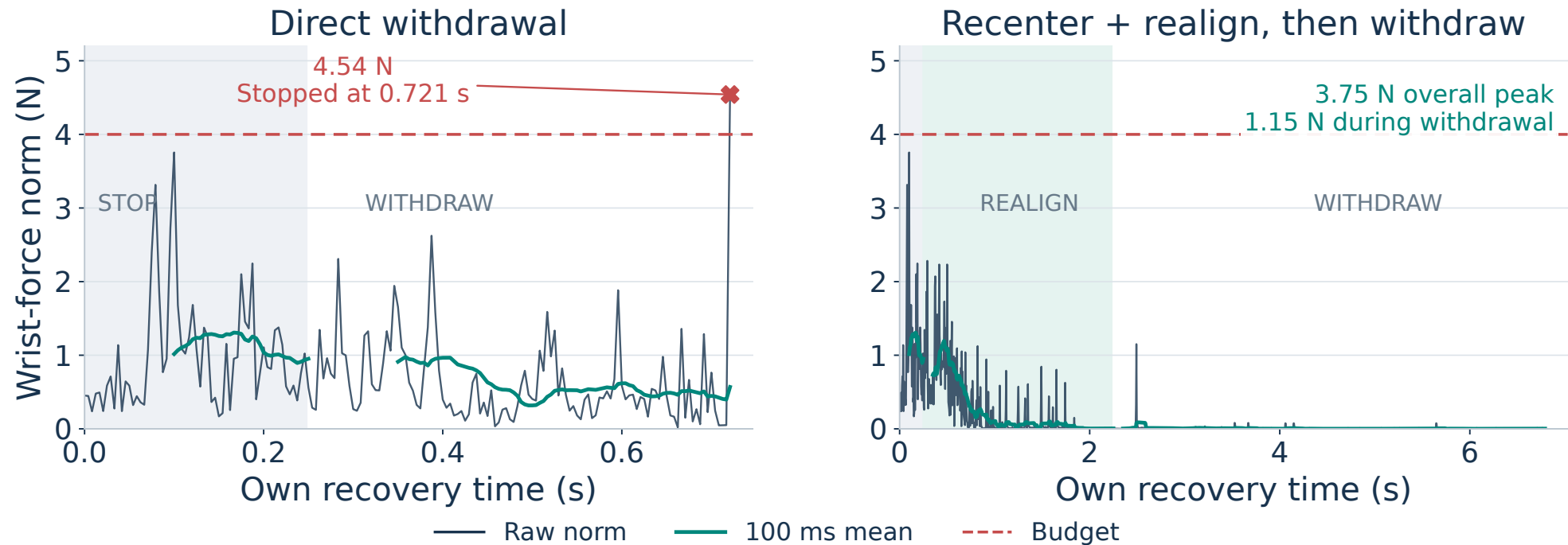
-3.009 mm depth, 0.063° tilt



Rigid bodies, 0.2 mm radial gap, 25 mm blind hole. Physics and task control: 240 Hz.

# Evidence: the exit depends on the motion

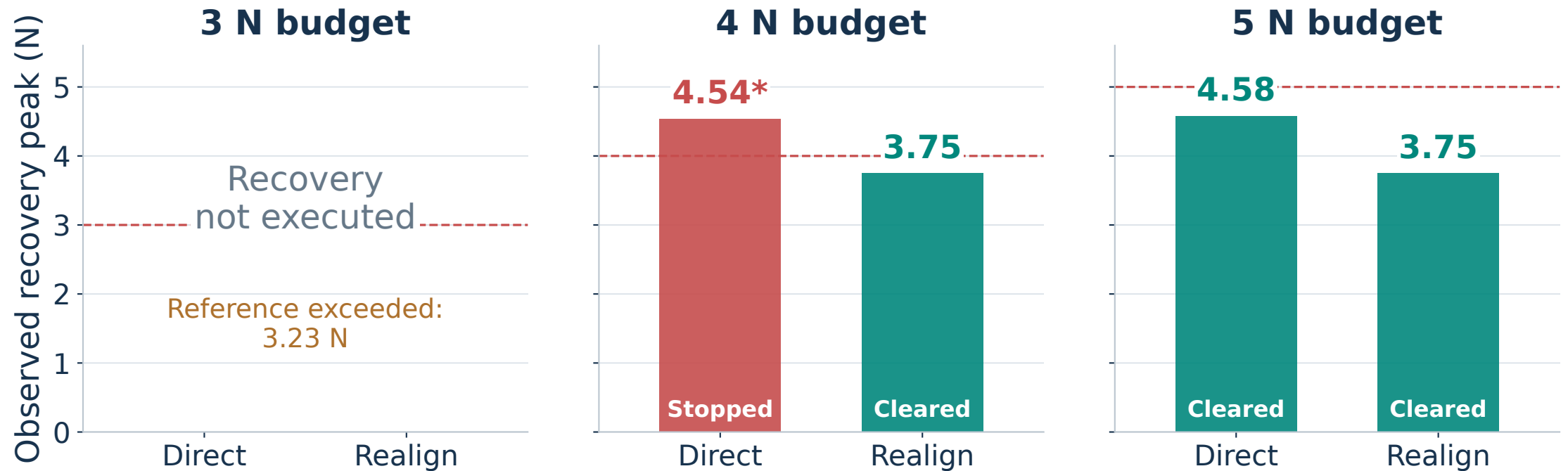
18 mm target depth ·  $\mu = 1$  ·  $1.5^\circ$  command tilt · 4 N budget



**Matched reference: 2,811 samples per policy · reference peak 3.56 N.**

Direct path stops at first exceedance. This pair motivates prevention; it does not validate it.

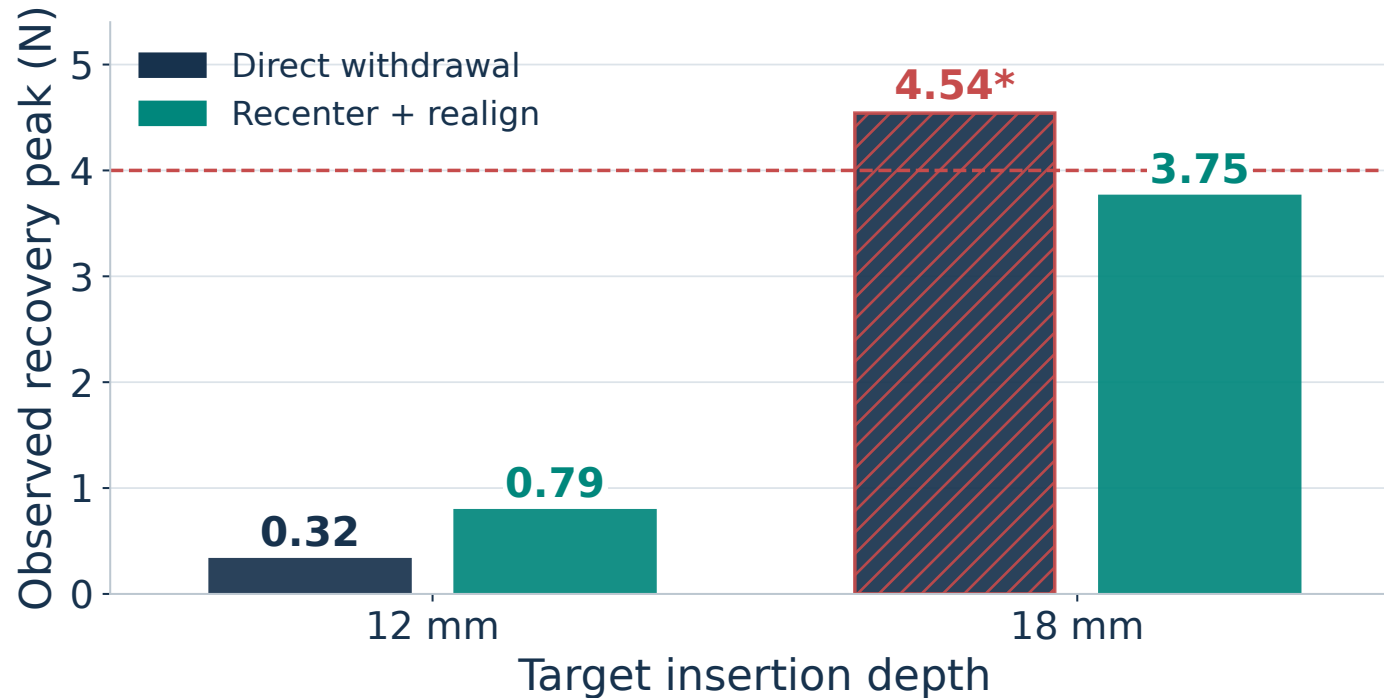
# The budget changes feasibility



\* First-exceedance stop; complete direct-withdrawal peak remains unobserved.

**Recoverability depends on state, candidate policy and the task budget.**

# Embedding depth changes the recovery cost



\* Censored at budget exceedance. Actual initial tilt: 1.52° / 1.44°.

**18 mm direct path**

**1.105 mm**

Commanded withdrawal  
at the stopping frame

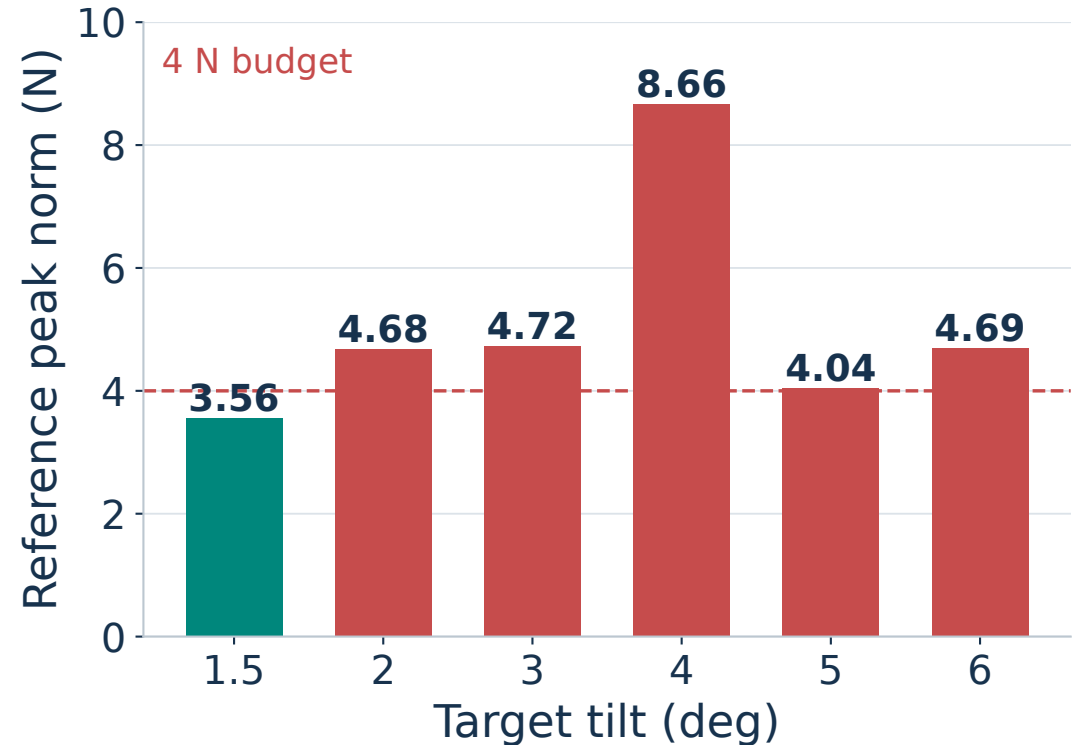
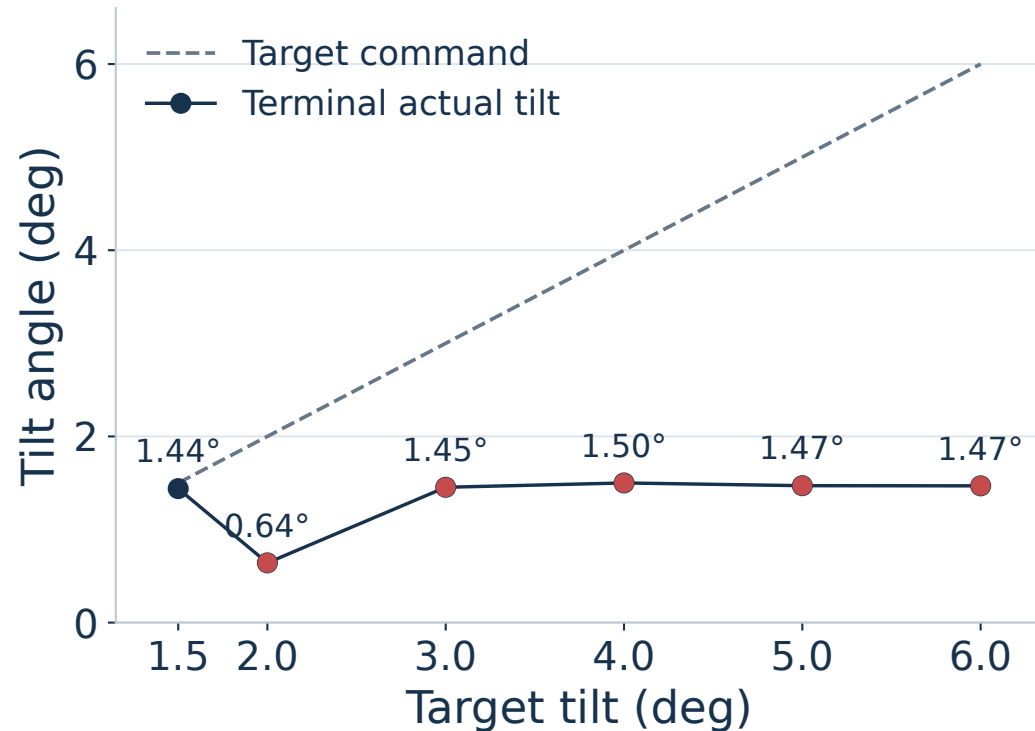
**0.137 mm**

Measured withdrawal  
at the same frame

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**Recovery cost can inform insertion risk. Prevention needs closed-loop tests.**

# A 6° command does not produce a 6° state

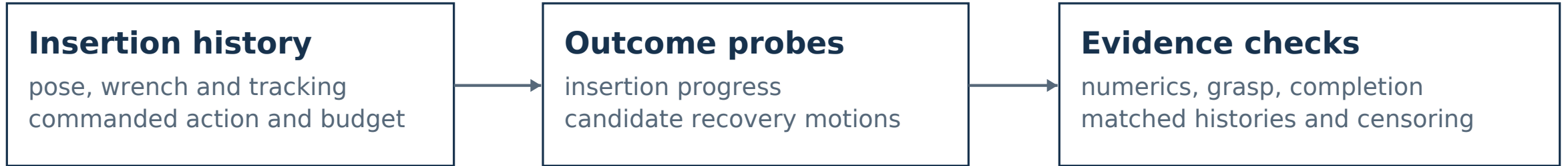


2°–6° commands: stopped during tilt, before withdrawal. Fixed 1 s ramp also changes angular speed.

**6° target: 3.743° last command, 1.469° actual tilt, 4.694 N at stopping.**

Actual state and force history are candidate risk inputs. No high-angle withdrawal was tested.

# A dataset for prevention and recovery



## 01 Insertion-risk targets

Persistent stall + elevated load  
Cross-check with an exit probe

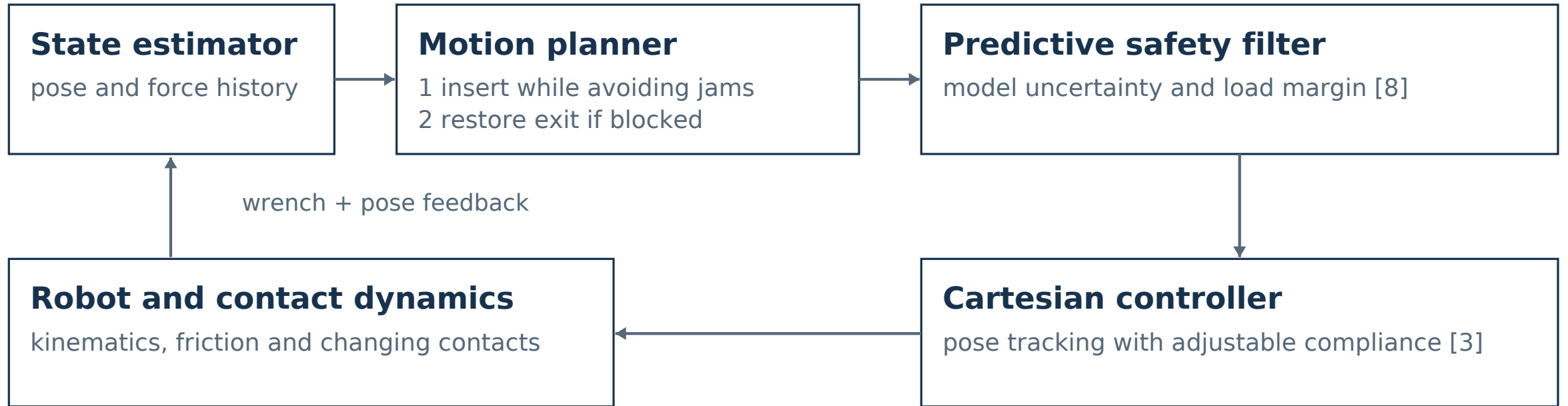
## 02 Recovery targets

Policy-specific exit outcome  
Peak load over adjustment and withdrawal

Risk indicators require validation. An over-budget probe is policy-specific.

**Group shared physical paths together when splitting training and test data.**

# A proposed planning and control loop



Hypothesis: pose and force history reveal insertion risk early enough to act.

Model predictive control (MPC) selects pose and speed under load and exit constraints.

Control barrier functions (CBFs) [5] require validated dynamics and feasible safety constraints.

# Ready to validate jam-avoiding insertion

## Demonstrated

Scenario: direct exit exceeds 4 N.  
 Pose adjustment enables an exit within budget.

### 01 Validate prevention

Compare baseline vs risk-aware insertion.  
 Measure completion, peak load and exit cost.

### 02 Restore withdrawability

Compare recovery paths from matched states.  
 Constrain the full adjustment-and-exit motion.

### 03 Robustness and transfer

Vary rates, contact settings and disturbances.  
 Calibrate grip and part limits before hardware trials.

Recovery feasibility shown. Jam-avoiding insertion remains to be validated.

# Complete evidence coverage

| Depth | μ   | Tilt | Budget | Direct withdrawal      | Recenter + realign  |
|-------|-----|------|--------|------------------------|---------------------|
| 18 mm | 1   | 0°   | 4 N    | Clear within budget    | Clear within budget |
| 18 mm | 1   | 1°   | 4 N    | Clear within budget    | Clear within budget |
| 18 mm | 1   | 1.5° | 4 N    | Exceeded in withdrawal | Clear within budget |
| 18 mm | 1   | 1.5° | 5 N    | Clear within budget    | Clear within budget |
| 12 mm | 1   | 1.5° | 4 N    | Clear within budget    | Clear within budget |
| 18 mm | 0.5 | 1.5° | 4 N    | Exceeded in reference  | Not tested          |
| 18 mm | 1   | 2°   | 4 N    | Exceeded in reference  | Not tested          |
| 18 mm | 1   | 1.5° | 3 N    | Exceeded in reference  | Not tested          |

[D1] Budget study: 8 conditions, 13 executed branches, 3 untested.

[D2] Angle extension: 6 conditions, 7 executed branches, 5 untested.

Exploratory deterministic trials. Shared histories and repeated anchors are not independent samples.

# References

## **[1] Whitney, D. E. (1982)**

Quasi-Static Assembly of Compliantly Supported Rigid Parts.

Journal of Dynamic Systems, Measurement, and Control 104(1):65-77. [DOI](#) / [paper](#)

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## **[2] Goli, F., et al. (2024)**

Characterizing the mechanics of rectangular peg-hole disassembly and the effect of the active compliance centre on the extraction force.

Royal Society Open Science 11:240956. [DOI](#) / [paper](#)

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## **[3] Hogan, N. (1985)**

Impedance Control: An Approach to Manipulation: Part I - Theory.

Journal of Dynamic Systems, Measurement, and Control 107(1):1-7. [DOI](#) / [paper](#)

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## **[4] Posa, M., Cantu, C., and Tedrake, R. (2014)**

A direct method for trajectory optimization of rigid bodies through contact.

International Journal of Robotics Research 33(1):69-81. [DOI](#) / [paper](#)

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# References continued

**[5] Ames, A. D., Xu, X., Grizzle, J. W., and Tabuada, P. (2017)**

Control Barrier Function Based Quadratic Programs for Safety Critical Systems.

IEEE Transactions on Automatic Control 62(8):3861-3876. [DOI](#) / [paper](#)

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**[6] Tang, B., et al. (2023)**

IndustReal: Transferring Contact-Rich Assembly Tasks from Simulation to Reality.

Robotics: Science and Systems XIX. [DOI](#) / [paper](#)

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**[7] Narang, Y., et al. (2022)**

Factory: Fast Contact for Robotic Assembly.

Robotics: Science and Systems XVIII. [DOI](#) / [paper](#)

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**[8] Wabersich, K. P., and Zeilinger, M. N. (2021)**

A predictive safety filter for learning-based control of constrained nonlinear dynamical systems.

Automatica 129:109597. [DOI](#) / [paper](#)

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